

Section 18 — System-Level Guarantees (Final Closure)

18.1 Introduction — From Components to a Unified System

Following:

- Section 16 — Validation
- Section 17A — Control
- Section 17B — Execution
- Section 17C — Protection

the system is now:

**Closed, Validated, Controlled, Executable,
and Protected**

18.2 Unified System State

$$S_k = (\Psi_k, P_k, PAC_k, D_k, \gamma_k, u_k)$$

with:

$$\mathcal{S}_k \in \mathcal{X}_{\text{safe}}$$

18.3 Global Evolution Law

$$\mathcal{S}_{k+1} = F_{\text{global}}(\mathcal{S}_k)$$

where:

$$\begin{aligned} F_{\text{global}} = & \\ & F_{\text{dyn}} \\ & \circ F_{\text{PAC}} \\ & \circ F_{\text{decision}} \\ & \circ F_{\text{validation}} \\ & \circ F_{\text{control}} \\ & \circ F_{\text{execution}} \end{aligned}$$

18.3' (Important Correction — Causality Order)

To ensure physical causality:

$$\begin{aligned} F_{\text{global}} = & \\ & F_{\text{dyn}} \\ & \circ F_{\text{prob}} \end{aligned}$$

$\circ F_{\mathrm{PAC}}$
 $\circ F_{\mathrm{validation}}$
 $\circ F_{\mathrm{decision}}$
 $\circ F_{\mathrm{control}}$
 $\circ F_{\mathrm{execution}}$

18.4 Closed-Loop Property

$\Psi \rightarrow P \rightarrow \text{PAC} \rightarrow D$
 $\rightarrow \gamma \rightarrow U \rightarrow \Psi$

$\rightarrow \text{No external decision pathway exists}$

18.5 Existence and Uniqueness

If:

- F_{global} is Lipschitz
- all inputs are bounded

then:

$\exists! \mathcal{S}_k$

18.6 Stability Guarantee

$$\begin{aligned} & \dot{\|\mathcal{L}\|}(\|\mathcal{S}\|) \\ & \leq \\ & - c\|\Psi\|^2 + C\sigma^2 \end{aligned}$$

with:

$$\gamma_k, u_k \rightarrow \text{no destabilizing control}$$

18.7 Safety Guarantee (Hard Gate)

$$\begin{aligned} & l_{\mathrm{eff}} < \theta_{\mathrm{ind}} \\ & \rightarrow \\ & \gamma_k = 0 \\ & \rightarrow \\ & u_k = 0 \end{aligned}$$

$$\rightarrow \text{No unsafe execution}$$

18.8 Adversarial Robustness Guarantee

- PAC fusion is median-based (breakdown = 33\%)
- Validation blocks invalid decisions
- $\mathcal{H}_\infty$ -type bounds limit disturbance impact

$\Rightarrow$ \text{Multi-layer adversarial protection}

18.9 Constraint Satisfaction

From primal–dual dynamics:

$$\Psi_k \in \Omega \quad \forall k$$

$\Rightarrow$ \text{Feasibility is invariant}

18.10 Optimality Guarantee

$$U^*(t) = \arg\min_{U \in \mathcal{U}_{\text{safe}}} \big(E(X) + \lambda J(U) \big)$$

$\Rightarrow$ \text{Optimal control under safety}

constraints}

18.11 Physical Realizability

- bounded delay
- bounded noise
- bounded actuation

$\Rightarrow$ \text{System is physically implementable}

18.12 Theorem — Global Validated Stability and Safety

Assume:

- $F_{\{\mathrm{global}\}}$ is Lipschitz
- PAC operator is bounded and contractive
- $I_{\{\mathrm{eff}\}} \geq$

$\theta_{\{\mathrm{ind}\}}$ at execution

- control is bounded
 - delay and noise are bounded
-

Then:

$\mathcal{S}_k \rightarrow \mathcal{S}^*$

and

the system remains stable, safe, and physically realizable

Proof (Sketch)

- From Section 17A: PAC induces convergence
- From Section 17B: physical system is stable
- From Section 17C: invalid inputs are rejected
- From Section 16: control is Lyapunov-safe

Thus:

$\dot{\mathcal{L}} \leq 0$
 $\rightarrow$ bounded trajectories
 $\rightarrow$ convergence

□

18.13 Non-Deception Property

Invalid data
 $\rightarrow$
 $\gamma = 0$
 $\rightarrow$
 $u = 0$
 $\rightarrow$
 No system impact

18.14 Adaptivity

- PAC updates dynamically
- weights evolve
- control adapts

$\rightarrow \text{Self-adaptive closed-loop system}$

18.15 Final Characterization

SEKHEM-TSDD is a:

- Closed system
- Stable system

- Safe system
- Adversarially robust system
- Physically executable system
- Adaptive system

FINAL PUBLISH-READY STATEMENT

`\boxed{`
`\text{SEKHEM-TSDD constitutes a fully closed,`
`validated, and adversarially robust decision-`
`control system, ensuring that only independent,`
`consistent, and physically feasible actions are`
`executed.}`
`}`

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
DOI: 10.5281/zenodo.19521219